

Aravindhan B

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PROFESSIONAL SUMMARY

AI/Product Engineer with 2+ years building 0-to-1 AI products and developer-facing systems across MCP, LLM agents, RAG, tool/function calling, Python/FastAPI, TypeScript/Node.js, Next.js/React, PostgreSQL, GraphQL, and cloud deployment. Hands-on with MCP-based agent/tool integrations, stateful workflows, API orchestration, authentication/permissions, observability, and production failure handling. Comfortable owning architecture through UX/DX, backend, frontend, and deployment in fast-moving startup environments.

WORK EXPERIENCE

Nuclei Tech Solutions - AI Engineer

Freelance(Dec 2025 - Present)

- Own embedded-device data acquisition, cleaning, and preprocessing pipelines for local, on-premise ERP systems, connecting device telemetry, application services, and enterprise data workflows where cloud dependence is constrained.
- Developed a lightweight MLP-Mixer-based architecture for resource-conscious AI workloads, designed to extend across multiple data modalities while preserving a compact footprint suitable for local enterprise deployment.

Loam AI - AI Engineer

Feb 2026 - June 2026

- Built production GenAI applications spanning LLMs, RAG, vector retrieval, tool-use agents, FastAPI services, and streaming React/React Native interfaces, owning features from backend integration to user-facing delivery.
- Developed reusable agent/tool orchestration components with structured schemas, state handling, API integrations, and retrieval pipelines so AI capabilities could support multiple product workflows instead of one-off demos.

Perspectiv Labs - Software Developer

July 2024 - Feb 2026

- Architected a modular full-stack platform using React/Next.js, TypeScript, PostgreSQL, GraphQL/REST, and Drizzle ORM, separating UI, service, and data layers for maintainable product development.
- Shipped 6+ production capabilities spanning RBAC authentication, email workflows, onboarding tours, currency-conversion markups, PWA push notifications, and CI/CD pipelines.
- Owned reusable frontend components, backend services, database models, and third-party integrations end-to-end, translating product requirements into maintainable production systems.

PROJECTS

CIEAV - Governed Agent Execution Platform | MCP, Python, TypeScript, LLMs, LangGraph

- Architected an open-source execution layer between AI agents and real-world tools, with MCP-based tool integration, capability discovery, scoped authority, and stateful orchestration across browser, API, and filesystem workflows.
- Implemented replay protection, idempotent effect handling, audit trails, and independent outcome observers that classify actions as PROVEN, DISPROVEN, or UNKNOWN instead of blindly retrying uncertain operations.

LWM OS - Agentic AI System | Python, TypeScript, MCP, PyTorch, LLMs, SQLite

- Built MCP discovery and tool orchestration across memory, dynamic DAG execution, approvals, telemetry, and workflow state, enabling agents to choose capabilities and execute multi-step tasks.
- Designed 21 capability blueprints and a sub-second planning path, focusing on developer control, inspectability, and reliable execution rather than opaque agent behavior.

Hallucination-Resistant LLM | RAG, Searx, Scrapy, LoRA/PEFT, Vector Search

- Built an open-source search-to-answer pipeline combining web collection, retrieval, instruction-tuned LLMs, and an entailment-based verifier that checks generated answers against retrieved evidence.
- Added dense vector retrieval, ASR/STT input, and containerized retrieval/generation/verification services with Docker/Kubernetes for repeatable deployment and evaluation.

TECHNICAL SKILLS

AI, Agents & MCP: Python, TypeScript, LLMs, MCP client/server integrations, tool/function calling, LangChain, LangGraph, CrewAI, multi-agent systems, RAG, embeddings, vector search, prompt engineering, evaluation, PyTorch, Transformers

Product, Backend & Data: FastAPI, Node.js, REST APIs, GraphQL, WebSockets/SSE, PostgreSQL, MySQL, SQLite, Redis, Supabase, FAISS, pgvector, OpenSearch, authentication/RBAC, API orchestration

Frontend & Infrastructure: Next.js, React, React Native (Expo), Tailwind CSS, Redux, Zustand, AWS, GCP, Docker, Kubernetes, CI/CD, Git

PUBLICATIONS

- Utilize AI Chatbots and Machine Learning To Address Depression, Attrition, and Follow-up Challenges - Aravindhan B
- Comprehensive Hallucination-Resistant Transformer Architecture with Advanced Neural Verification, Multi-Question Batch Processing and Cloud Deployment - Aravindhan B

EDUCATION

BE - Computer Science Engineering,
Chettinad College of Engineering and Technology

Karur, Tamil Nadu
Aug 2020 - Jun 2024